

THISAL DILMITH GONSALKORALA

Software Engineer

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SUMMARY

Software Engineer specializing in Python, FastAPI, Spring Boot, Node.js, React, and Next.js, with experience building scalable web applications, AI-powered systems, and distributed microservices. Recently completed a **B.Sc. (Hons) in Information Technology (Information Technology)** at SLIIT, with graduation scheduled for next month. Proven leadership in delivering an AI-powered research platform from architecture to deployment.

EXPERIENCE

Associate Software Engineer — ABMSZ

Aug 2024 – Present

- Supervise and mentor a team of interns, reviewing code and guiding delivery on production features.
- Build and maintain full-stack MERN/TypeScript applications, including authentication, RBAC, and data layer design.
- Implement production security controls (field-level AES-256-GCM encryption, JWT rotation with reuse detection) reducing exposure of sensitive data in transit and at rest.
- Own deployment pipelines on Vercel, resolving CORS, environment configuration, and connection-pooling issues to keep services stable in production.

SELECTED PROJECTS

EduSense — Neuro-Adaptive Learning Platform (Uni Research) — Python, Random Forest, LLM/RAG

- Led a four-member team and developed the **Cognitive-Linguistic Adaptation Engine**, implementing a **Random Forest** cognitive load prediction model and an **LLM/RAG-powered** adaptive content generation pipeline within a distributed microservice architecture.
- Designed and integrated REST APIs, curriculum-aware validation, and AI-driven lesson adaptation using **KPR** and **Complexity Reduction (CR)** metrics. Delivered **cognitive state-based personalized lessons**, AI-generated visual learning, and synchronized **audio, visual, and haptic feedback**. Validated the platform with **50 Grade 6 students**, achieving **89.39% cognitive load classification accuracy**.

InventoryMate (ULSS Fleet & Inventory Management) — MERN, SheetJS, Node.js v22

- Shipped a live production admin platform (inventorymate.vercel.app) managing vehicles, spare parts, technicians, and maintenance records for a fleet operator.
- Built bulk CSV/Excel import with SheetJS, fixing numeric-parsing edge cases to keep inventory data integrity at scale.
- Debugged and resolved production issues spanning CORS, MongoDB connection pooling, IPv6/DNS resolution, and SMTP integration for demo-request workflows.

Gevora — Sri Lankan Property Portal — Next.js, TypeScript, Fastify, PostgreSQL/PostGIS, Prisma, Elasticsearch, Redis, BullMQ, Turborepo

- **Architected and developed** a full-stack real estate platform using a **Turborepo** microservice architecture, implementing secure authentication, geospatial property search, high-performance REST APIs, and scalable backend services.
- **Engineered** a hybrid search architecture with **PostgreSQL/PostGIS** and **Elasticsearch**, designing separate property and listing domains, background job processing with **BullMQ**, and Redis caching to optimize search performance and system scalability.

EDUCATION

BSc (Hons) Information Technology, Specialization in Information Technology

Expected 2026

Sri Lanka Institute of Information Technology (SLIIT) — Final Year

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, Kotlin
- **Frontend:** React, React Native (Expo), Next.js, Tailwind CSS, Chart.js
- **Backend:** Node.js, Express.js, Spring Boot, Fastify
- **Databases:** MongoDB, PostgreSQL (PostGIS), Redis, Elasticsearch
- **Cloud & DevOps:** AWS (Lambda, S3, DynamoDB, API Gateway), Docker, GitHub Actions, Terraform, Vercel deployment
- **Security:** AES-256-GCM encryption, JWT rotation & reuse detection, RBAC, Zod validation
- **Infrastructure & Architecture:** Socket.io, BullMQ, Turborepo monorepos
- **Data Science / ML:** scikit-learn, data preprocessing, model deployment via REST API
- **Testing & QA:** Selenium, TestNG, PyTest, Testcontainers (JUnit5)
- **Practices:** Team leadership, code review, mentoring interns, agile delivery (Jira/Trello), sprint management